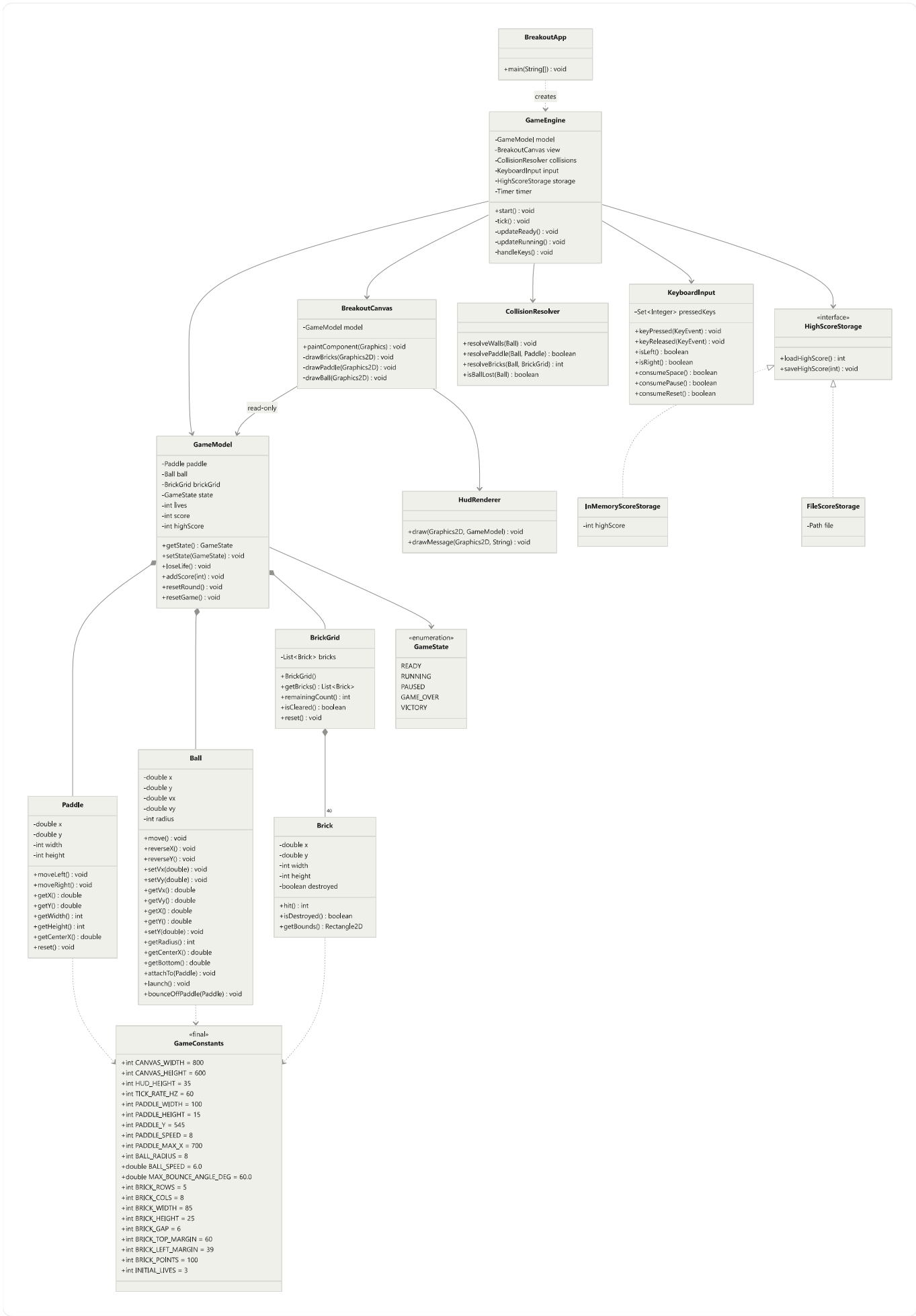
# نمودار کلاس Breakout (فاز ۱)

این فایل Mermaid است و GitHub آن را به‌صورت خودکار رسم می‌کند.


BreakoutApp
+main(String[]) : void
creates
GameEngine
-GameModel model
-BreakoutCanvas view
-CollisionResolver collisions
-KeyboardInput input
-HighScoreStorage storage
-Timer timer
+start() : void
-tick() : void
-updateReady() : void
-updateRunning() : void
-handleKeys() : void
BreakoutCanvas
-GameModel model
+paintComponent(Graphics) : void
-drawBricks(Graphics2D) : void
-drawPaddle(Graphics2D) : void
-drawBall(Graphics2D) : void
CollisionResolver
+resolveWalls(Ball) : void
+resolvePaddle(Ball, Paddle) : boolean
+resolveBricks(Ball, BrickGrid) : int
+isBallLost(Ball) : boolean
KeyboardInput
-Set<Integer> pressedKeys
+keyPressed(KeyEvent) : void
+keyReleased(KeyEvent) : void
+isLeft() : boolean
+isRight() : boolean
+consumeSpace() : boolean
+consumePause() : boolean
+consumeReset() : boolean
«interface»
HighScoreStorage
+loadHighScore() : int
+saveHighScore(int) : void
read-only
GameModel
-Paddle paddle
-Ball ball
-BrickGrid brickGrid
-GameState state
-int lives
-int score
-int highScore
+getState() : GameState
+setState(GameState) : void
+loseLife() : void
+addScore(int) : void
+resetRound() : void
+resetGame() : void
HudRenderer
+draw(Graphics2D, GameModel) : void
+drawMessage(Graphics2D, String) : void
InMemoryScoreStorage
-int highScore
FileScoreStorage
-Path file
BrickGrid
-List<Brick> bricks
+BrickGrid()
+getBricks() : List<Brick>
+remainingCount() : int
+isCleared() : boolean
+reset() : void
«enumeration»
GameState
READY
RUNNING
PAUSED
GAME_OVER
VICTORY
Paddle
-double x
-double y
-int width
-int height
+moveLeft() : void
+moveRight() : void
+getX() : double
+getY() : double
+getWidth() : int
+getHeight() : int
+getCenterX() : double
+reset() : void
Ball
-double x
-double y
-double vx
-double vy
-int radius
+move() : void
+reverseX() : void
+reverseY() : void
+setVx(double) : void
+setVy(double) : void
+getVx() : double
+getVy() : double
+getX() : double
+getY() : double
+setY(double) : void
+getRadius() : int
+getCenterX() : double
+getBottom() : double
+attachTo(Paddle) : void
+launch() : void
+bounceOffPaddle(Paddle) : void
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Brick
-double x
-double y
-int width
-int height
-boolean destroyed
+hit() : int
+isDestroyed() : boolean
+getBounds() : Rectangle2D
«final»
GameConstants
+int CANVAS_WIDTH = 800
+int CANVAS_HEIGHT = 600
+int HUD_HEIGHT = 35
+int TICK_RATE_HZ = 60
+int PADDLE_WIDTH = 100
+int PADDLE_HEIGHT = 15
+int PADDLE_Y = 545
+int PADDLE_SPEED = 8
+int PADDLE_MAX_X = 700
+int BALL_RADIUS = 8
+double BALL_SPEED = 6.0
+double MAX_BOUNCE_ANGLE_DEG = 60.0
+int BRICK_ROWS = 5
+int BRICK_COLS = 8
+int BRICK_WIDTH = 85
+int BRICK_HEIGHT = 25
+int BRICK_GAP = 6
+int BRICK_TOP_MARGIN = 60
+int BRICK_LEFT_MARGIN = 39
+int BRICK_POINTS = 100
+int INITIAL_LIVES = 3

## لایه‌ها

| لایه | کلاس‌ها |
|---|---|
| Model | `GameConstants` , `GameState` , `GameModel` , `BrickGrid` , `Brick` , `Ball` , `Paddle` |
| View | `HudRenderer` , `BreakoutCanvas` |
| Controller | `KeyboardInput` , `CollisionResolver` , `GameEngine` |
| Storage | `FileScoreStorage` , `InMemoryScoreStorage` , `HighScoreStorage` |

## State machine

